# Treasury System Structure for Thai Commercial Bank: Local Bond and Money Market Operations

## 1. System Overview and Strategic Objectives

### 1.1 Primary System Goals

**1.1.1 Liquidity Management Function** The **liquidity management function** serves as the foundational pillar of the treasury system, enabling precise control over short-term funding needs and strategic deployment of excess cash in the Thai Baht market. This function operates through two interconnected channels: **bilateral repos with the Bank of Thailand** for monetary policy operations, and **private market repos** with interbank counterparties for flexible short-term funding management.

The Thai repo market has evolved substantially since the Bank of Thailand discontinued its centralized repo operations in 2008, transitioning to a bilateral framework that has driven significant market growth. By 2013, private repo market volume had grown to more than **twice the size of the interbank unsecured market**, with bilateral repos accounting for approximately **85% (THB 166 trillion)** of total market volume while private repos comprised **15% (THB 28 trillion)**. This structural transformation necessitates sophisticated liquidity forecasting tools that model both secured and unsecured funding scenarios, with particular attention to **collateral availability** and **haircut requirements** that govern market participation.

The system must implement **real-time cash position monitoring** across all nostro accounts, with particular emphasis on BAHTNET settlement accounts maintained with the Bank of Thailand. **BAHTNET**, Thailand's real-time gross settlement system, serves as the backbone for high-value interbank payments and securities settlement, providing **immediate finality** that eliminates settlement risk for time-critical operations. The liquidity forecasting module must incorporate multiple data sources including maturing investments, scheduled repo expiries, anticipated deposit flows, and seasonal patterns specific to Thai corporate behavior—particularly around **month-end, quarter-end, and Thai public holidays** when market liquidity may be constrained.

A critical component is the **Thai Overnight Repurchase Rate (THOR)**, which has emerged as the key benchmark for interbank overnight private repo transactions. Administered by the Bank of Thailand with ThaiBMA as calculation agent, THOR must be embedded in the system's pricing engines for accurate repo rate determination and relative value analysis. The system should also accommodate **THOR-based floating rate instruments** as they develop in the Thai market.

Repo transaction support must address **Thai-specific settlement conventions**, including typical value date structures of **"3 plus zero"** (3-day settlement for first leg, same day for second leg), **"2 plus 1,"** or **"2 plus 2"** for repo transactions, with tenors typically extending up to **1 month** but customizable based on counterparty agreements. The system must automatically calculate settlement dates, track pending cash flows, and provide **early warning alerts** for potential liquidity shortfalls or surpluses requiring market intervention.

**1.1.2 Investment Portfolio Management** The **investment portfolio management function** focuses on **optimizing risk-adjusted returns** within defined parameters across the spectrum of THB-denominated fixed income instruments. Given the **25 transaction/day volume ceiling**, the system prioritizes **analytical sophistication and control robustness** over high-frequency trading capabilities, emphasizing **well-researched, compliance-vetted investment decisions** aligned with the bank's risk appetite.

The investment universe spans: **Thailand government bonds** (T-bonds and T-bills), **Bank of Thailand securities**, **state-owned enterprise bonds**, **corporate bonds**, and **structured repo transactions**. The **seamless sovereign yield curve** created through coordination between the Ministry of Finance and Bank of Thailand requires particular attention—the BOT occupies the **sub-three-year segment** with its bond issuance, while the Ministry of Finance takes **longer tenors**, resulting in instruments that trade in an almost seamless manner despite different issuers.

| Instrument Category | Typical Maturity | Primary Purpose | Settlement Infrastructure | Key Market Characteristic |
|---|---|---|---|---|
| **T-Bills** | 91–364 days | Liquidity buffer, collateral | BAHTNET RTGS | Discount instruments, auction-issued |
| **T-Bonds** | 2–50+ years | Investment, collateral, HQLA | BAHTNET RTGS | Benchmark yield curve, semi-annual coupons |
| **BOT Bonds/ Bills** | 1–2 years (bonds); <1 year (bills) | Monetary policy operations, liquidity management | BAHTNET RTGS | Policy rate linkage, repo eligibility |
| **SOE Bonds** | 3–15 years | Yield enhancement, quasi-sovereign exposure | TSD DVP Model 1 | Government guarantee (typical), spread to sovereign |
| **Corporate Bonds** | 1–10 years | Credit spread income | TSD DVP Model 1 | Rating-diverse, liquidity varies by issuer |
| **Repos/Reverse Repos** | Overnight–1 month | Short-term funding/cash deployment | BAHTNET/TSD | Collateralized, haircut-adjusted |

Portfolio construction requires **strategic asset allocation** tools that set target ranges for exposure to different instrument types, credit qualities, and maturity buckets. The system must incorporate **liquidity-adjusted risk measures** that penalize concentrations in thinly traded issues, particularly relevant for **corporate bonds where secondary market liquidity can be constrained**. **Relative value analytics** should identify cheap or rich sectors based on historical yield relationships, with particular attention to **SOE bond spreads to government benchmarks** and **corporate credit curve positioning**.

Security selection integrates **credit analysis capabilities** including financial statement analysis with Thai accounting standard-specific metrics, **industry benchmarking** against Thai and regional peers, and **macroeconomic scenario analysis** relevant to Thailand's economic cycles. For **corporate bonds**, the system must track **SEC-recognized credit ratings** (Fitch, Moody's, S&P, TRIS Rating) and support **internal credit assessments** with documented methodologies.

#### 1.1.3 Local THB Market Focus

The **exclusive THB focus** eliminates multi-currency complexity while demanding **deep specialization in Thai market infrastructure and regulatory requirements**. This specialization manifests in several critical design decisions:

**First**, the system must maintain **native Thai baht currency handling** with proper formatting, rounding conventions, and **regulatory reporting formats prescribed by the Bank of Thailand**. **Second**, integration with Thai market infrastructure—**TSD for corporate bond settlement**, **BAHTNET for government securities and cash settlement**, and **ThaiBMA for trade reporting**—must be architected as **core system capabilities** rather than peripheral add-ons. **Third**, compliance with **Thai Financial Reporting Standards (TFRS)**, which align with **IFRS 9** for financial instrument accounting, must be embedded in the accounting engine.

The **overwhelming dominance of OTC trading**—approximately **99% of bond secondary market volume versus 1% on the Thailand Bond Exchange (TBX)**—shapes trade execution module design toward **voice-brokered and bilateral negotiation workflows** rather than electronic order book functionality. Similarly, the **30-minute ThaiBMA trade reporting obligation**necessitates **robust trade capture and automated reporting capabilities** with built-in validation for completeness and accuracy.

### 1.2 Target Operating Environment

#### 1.2.1 Low-Volume Transaction Profile ( 25 transactions/day)

The **25 transaction/day ceiling** fundamentally shapes architectural decisions toward **reliability, auditability, and analytical sophistication** over raw throughput. At this volume—approximately **6,250 transactions annually** assuming 250 trading days—the system can employ:

| Architectural Element | Low-Volume Optimization | Typical Implementation |
|---|---|---|
| **Database** | ACID compliance over horizontal scaling | Oracle/PostgreSQL/SQL Server with full normalization |
| **Processing model** | Synchronous with immediate commit | Real-time validation, multi-level approval workflows |
| **User interface** | Rich contextual data display | Comprehensive counterparty info, historical patterns, risk analytics |
| **Batch windows** | Generous timeframes for thorough validation | Extended EOD processing, complex calculations without time pressure |
| **Infrastructure** | Conservative sizing with headroom | Single application server with redundant failover |

The low-volume environment enables **"white glove" transaction handling** where each trade receives **individual attention from qualified personnel**—pre-execution limit checking, post-execution confirmation matching, and settlement instruction verification with manual oversight at each stage. However, the architecture must accommodate **burst capacity** for: month-end/quarter-end portfolio rebalancing, emergency liquidity operations during market stress, simultaneous repo executions for overnight funding, and bulk processing of maturing positions.

**1.2.2 Multi-Layer User Segregation (Front/Middle/Back Office + IT Admin)** The **four-tier segregation** enforces **separation of duties** critical for operational risk management and regulatory compliance:

| Tier | Primary Functions | Key Permissions | Explicit Restrictions |
|---|---|---|---|
| **Front Office** | Trade execution, portfolio management, market analysis | Real-time market data, trade capture, position monitoring, P&L inquiry | No settlement instruction modification; no accounting record access; no user administration |
| **Middle Office** | Risk management, compliance monitoring, independent validation | Limit setting/ monitoring, regulatory reporting, price verification, exception investigation | No trade execution; no settlement processing |
| **Back Office** | Settlement, accounting, reconciliation | Trade confirmation, settlement instruction generation, journal entry preparation, nostro reconciliation | No trade execution; no limit modification |
| **IT Administration** | System configuration, user provisioning, technical support | Static data maintenance, security configuration, interface monitoring | No live transaction data access; no trade processing influence |

**Technical enforcement** through role-based access control (RBAC) at database and application levels prevents function consolidation that would create control vulnerabilities. **Four-eyes principle** implementation requires **mandatory secondary approval** for transactions exceeding thresholds, with **immutable logging** of all approval actions.

**1.2.3 Integration with Thai Financial Market Infrastructure** Effective treasury operations require **seamless, automated integration** with three critical external systems:

| Infrastructure | Function | Integration Requirements | Critical Features |
|---|---|---|---|
| **Thailand Securities Depository (TSD)** | Central securities depository for corporate bonds, equities; DVP settlement | SWIFT/proprietary messaging, PSMS matching, position inquiry, corporate actions | Scripless book-entry; omnibus/segregated accounts; T+2 settlement; no debit balances since Nov 2017 |
| **BAHTNET** | RTGS for government securities, high-value payments | Real-time payment initiation, liquidity monitoring, settlement finality confirmation | Immediate irrevocable settlement; systemic importance requiring redundant connectivity |
| **ThaiBMA** | Trade reporting, reference pricing, market surveillance | 30-minute trade reporting API, price feed ingestion, market data services | Regulatory-mandated timeliness; authoritative mark-to-market source |

## 2. Supported Instruments and Product Scope

### 2.1 Sovereign and Central Bank Instruments

**2.1.1 Thailand Government Bonds (T-Bonds, T-Bills)** **Thailand government bonds** constitute the **foundational risk-free asset class** for Thai commercial bank treasury operations. The **Public Debt Management Office (PDMO)** conducts regular auctions: **Treasury Bills on Mondays** and **Government Bonds (Loan Bonds) on Wednesdays**, both using **multiple-price auction formats**.

| Instrument | Maturity Range | Coupon Structure | Day Count | Settlement | Primary Market Access |
|---|---|---|---|---|---|
| **T-Bills** | 91, 182, 364 days | Zero-coupon, discount | Actual/365 | T+0 to T+2 via BAHTNET | Auction, primary dealers |
| **T-Bonds** | 2–50+ years | Fixed or floating, semi-annual typical | Actual/Actual or Actual/365 | T+2 via BAHTNET | Auction, primary dealers; secondary OTC |

**Accounting treatment** under TFRS 9 depends on business model: **amortized cost** for hold-to-collect liquidity buffers; **FVOCI** for flexible hold-to-collect-and-sell strategies; **FVTPL** for active trading positions. The system must support **effective interest rate calculation**, **premium/discount amortization**, and **impairment assessment** (though government bonds typically qualify for **low credit risk**

**exemption** under ECL).

**Collateral eligibility** for BOT operations makes government bonds the **preferred repo collateral**, with **haircuts ranging from 1% (0–5 years) to 5% (>20 years)**. The system must track **eligible collateral status** dynamically as the BOT updates its acceptable securities list.

**2.1.2 Bank of Thailand Bonds and Bills (BOT Bonds/Bills)** **BOT securities** serve **monetary policy implementation** and provide **additional risk-free investment options**. **BOT bonds** (1–2 years) and **BOT bills** (<1 year) are auctioned **on Tuesdays** using **multiple-price formats**. These instruments are **actively used in BOT bilateral repo operations**, which accounted for ~**85% of total repo market volume (THB 166 trillion)** in 2013.

| BOT Operation | Purpose | Typical Tenor | Rate Benchmark |
|---|---|---|---|
| Bilateral repo (bank borrows from BOT) | Liquidity injection to banking system | Overnight–1 month | Policy rate-linked |
| Bilateral reverse repo (bank lends to BOT) | Liquidity absorption, cash deployment | Overnight–1 month | Policy rate-linked |
| BOT bond/bill auction | Monetary policy signaling, yield curve influence | 1–2 years (bonds); <1 year (bills) | Market-determined auction |

The system must support **complete repo auction lifecycle**: announcement parsing, bid preparation with **collateral optimization** (considering haircuts and liquidity needs), submission, allocation processing, and **collateral transfer/settlement tracking**. **Haircut matrices** for BOT operations must be dynamically applied - | **Collateral Type** | **0–5 Years** | **5–10 Years** | **10–20 Years** | **>20 Years** | |:—|:—|:—|:—|:—| | Government bonds | 1% | 3% | 4% | 5% | | BOT bonds/bills | 1% | 3% | 4% | 5% | | SOE bonds (guaranteed) | 1% | 3% | 4% | 5% | | Corporate bonds (AAA) | 5% | 10% | 15% | 20% | | Corporate bonds (AA) | 10% | 15% | 20% | 25% | | Corporate bonds (A) | 15% | 20% | 25% | 30% | | Corporate bonds (BBB) | 20% | 25% | 30% | 35% |

### 2.2 Quasi-Sovereign Instruments

**2.2.1 State-Owned Enterprise Bonds (SOE Bonds)** **SOE bonds** represent **quasi-sovereign credit risk** with **implicit or explicit government guarantees**. Major issuers include **EGAT, MEA, PEA, State Railway of Thailand**, and other government-controlled entities. **SOE bond outstanding** was **THB 1,033,961.78 million** as of recent public debt statistics.

**Key characteristics** requiring system support:

| Feature | System Implication |
|---|---|
| **Government guarantee** (typical) | Track guarantee terms, claim procedures; sovereign-equivalent risk weight for capital |

| Feature | System Implication |
|---|---|
| **Single-price auction (Thursdays)** | Distinct auction workflow from multiple-price government bond auctions |
| **BOT registrar settlement** | BAHTNET integration similar to government bonds |
| **Preferential haircut treatment** | Same as government bonds: 1–5% by maturity |
| **Yield pickup over sovereign** | Relative value analytics, spread monitoring |

Despite government backing, the system must maintain **independent credit monitoring** of underlying SOE financial performance as **early warning indicators** for potential guarantee invocation scenarios.

**2.2.2 Government-Guaranteed Corporate Obligations** Certain **corporate obligations carry explicit government guarantees**, typically under **specific government programs** supporting strategic industries. The system must:

- **Bifurcate treatment**: guaranteed portion as sovereign risk; unguaranteed portion as corporate credit risk
- **Document guarantee terms**: coverage percentage, expiration, conditions precedent, claim procedures
- **Track substitution history** if collateral changes

### 2.3 Private Sector Debt Instruments

**2.3.1 Corporate Bonds (Investment Grade)** **Corporate bonds** provide **credit spread income** with **rating-diverse exposure**. The Thai corporate bond market has grown substantially, with **average annual issuances rising from 112 (2015–2019) to 139 (2020–2024)** and **outstanding reaching USD 166 billion by end-2024**.

| Issuance Channel | Regulatory Requirements | Typical Investors | System Support Needed |
|---|---|---|---|
| **Public offering** | SEC approval, full disclosure, investment-grade rating | Institutions, retail | Comprehensive disclosure tracking, rating integration |
| **PP-10** ( 10 investors, 4-month period) | No SEC approval | Sophisticated investors | Investor count tracking, transfer restriction monitoring |
| **PP-II** (institutional investors) | SEC approval, additional disclosure | Qualified institutions | Investor qualification verification |

| Issuance Channel | Regulatory Requirements | Typical Investors | System Support Needed |
|---|---|---|---|
| **PP-HNWI** (high net worth) | SEC approval, enhanced disclosure | HNWI individuals | Suitability documentation |

**TFRS 9 ECL requirements** are particularly significant for corporate bonds. The system must support:

| ECL Stage | Trigger | Measurement | System Functionality |
|---|---|---|---|
| **Stage 1** | No significant credit deterioration | 12-month ECL | PD/LGD/EAD estimation, forward-looking macro scenarios |
| **Stage 2** | Significant credit deterioration | Lifetime ECL | Deterioration indicators, migration tracking |
| **Stage 3** | Credit-impaired (objective evidence) | Lifetime ECL, net carrying amount interest | Impairment recognition, recovery analysis |

**Settlement** follows **T+2 DVP through TSD**, with **PSMS matching** requiring precise trade detail alignment.

**2.3.2 Commercial Paper and Short-Term Debt** **Commercial paper** ( 270 days, typically discount basis) serves **temporary liquidity deployment**. The system must support:

- **Accretion accounting** for discount instruments
- **Rollover risk monitoring**: projection of maturity concentrations, reinvestment planning
- **Tighter limits**: single issuer, sector concentration controls given shorter-term credit sensitivity

### 2.4 Repurchase Agreement Instruments

**2.4.1 Bilateral Repos with Bank of Thailand (Monetary Policy Operations)** **BOT bilateral repos** are **restricted to appointed primary dealers** and serve as the **principal monetary policy transmission channel**. The system must implement:

| Process Stage | System Functionality |
|---|---|
| **Announcement receipt** | Parse BOT operation parameters (tenor, amount, eligible collateral) |

| Process Stage | System Functionality |
|---|---|
| **Bid preparation** | Optimize collateral selection considering haircuts, liquidity needs, portfolio impact |
| **Bid submission** | Format and transmit bids in BOT-prescribed format |
| **Allocation processing** | Record successful awards, update positions |
| **Collateral transfer** | Generate BAHTNET securities transfer instructions |
| **Term management** | Daily mark-to-market, margin monitoring, interest accrual |
| **Maturity/rollover** | Cash repayment planning, new bid preparation if rolling |

**2.4.2 Private Market Repos (Interbank and Institutional)** **Private repos** provide **flexible short-term funding** beyond BOT operations, governed by **GMRA with Thailand Annex** or **Thai Master Repurchase Agreement**. **Market data from 2016** showed **overnight repos at 37% of outstanding volume**, with **tenors of one month or longer at 14%**, and **total outstanding at THB 475 billion**.

| Repo Structure | Characteristics | System Requirements |
|---|---|---|
| **Classic repo** | Specific collateral, fixed term | Collateral identification, substitution tracking |
| **Open repo** | No fixed maturity, callable | Daily rate reset, continuous collateral management |
| **General collateral (GC) repo** | Collateral pool, rate-focused | Pool management, optimization algorithms |
| **Special repo** | Specific security in demand | Specials tracking, rate inversion monitoring |

**Collateral management** must implement **variation margin in cash form with interest inclusion**—a Thai market convention. **Margin call threshold: THB 500,000**, with automated calculation:

$$\text{Margin Call} = \text{Market Value} - \frac{\text{Purchase Price} + \text{Accrued Interest}}{1 - \text{Initial Margin}}$$

**2.4.3 Reverse Repos for Liquidity Deployment** **Reverse repos** (cash lending against collateral) require **enhanced credit risk assessment**:

| Risk Element | Mitigation | System Implementation |
|---|---|---|
| Counterparty default | Collateral liquidation | Collateral eligibility, haircut adequacy, liquidation procedures |
| Collateral value decline | Margin calls, haircuts | Daily mark-to-market, automated margin call generation |
| Collateral illiquidity | Concentration limits, liquidity buffers | Collateral quality scoring, stress testing |

**2.4.4 GMRA and Thai Master Repurchase Agreement Compliance** The system must maintain **master agreement repository** with:

| Tracked Elements | Purpose |
|---|---|
| Agreement version and amendments | Ensure current terms apply |
| Counterparty coverage | Validate transaction eligibility |
| Margin maintenance thresholds | Trigger automated margin processes |
| Collateral eligibility schedules | Filter available securities |
| Haircut matrices | Calculate required collateral |
| Events of default and close-out netting | Support default procedures |

## 3. Core Functional Modules

### 3.1 Trade Execution Module

**3.1.1 OTC Trade Capture and Negotiation** Given ~**99% OTC market share**, the module prioritizes **voice-brokered transaction capture** with:

| Capture Element | Validation | Enrichment |
|---|---|---|
| Instrument (ISIN/internal code) | Master data existence, eligibility | Description, rating, maturity, coupon |
| Trade date/time | Business day, market hours | Timestamp for 30-min reporting |
| Settlement date | T+2 standard, holiday adjustment | Automatic calculation with override |

| Capture Element | Validation | Enrichment |
|---|---|---|
| Price/yield | Reasonableness range | Clean/dirty price conversion, accrued interest |
| Notional amount | Limit pre-check | Unit conversion (THB 1,000/100,000 typical) |
| Counterparty | KYC status, limit availability | Historical relationship data |
| Repo terms (if applicable) | Collateral eligibility, rate reasonableness | Haircut application, maturity date |

**Negotiation support** includes **quote logging** for best execution documentation and **work-up tracking** from indicative to firm prices.

#### 3.1.2 Order Management and Workflow Routing

| Workflow State | Responsible Party | Action Required | Escalation Trigger |
|---|---|---|---|
| Created | Front office trader | Entry completion | — |
| Pending approval | Front office supervisor | Four-eyes confirmation | Amount threshold, non-standard terms |
| Approved | System | Limit check, validation | Limit breach |
| Executed | Counterparty | Trade confirmation | Confirmation mismatch |
| Confirmed | Middle office | Independent verification | Price anomaly, unusual pattern |
| Settled | Back office | Instruction generation, transmission | Settlement failure |

#### 3.1.3 Counterparty Limit Monitoring and Pre-Trade Controls

**Real-time limit framework**:

| Limit Type | Calculation Basis | Response to Breach |
|---|---|---|
| Single transaction | Notional amount | Hard block with override workflow |
| Counterparty aggregate | Sum of exposures (settled + pending) | Warning at 80%, block at 100% |

| Limit Type | Calculation Basis | Response to Breach |
|---|---|---|
| Tenor-based | Exposure by maturity bucket | Adjusted for collateral, netting |
| Concentration | Issuer, sector, rating distribution | Hard caps with board exception |
| Repo-specific | Cash amount vs. haircut-adjusted collateral | Dynamic based on collateral quality |

#### 3.1.4 Price Discovery and Market Data Integration

| Data Source | Content | Frequency | Use Case |
|---|---|---|---|
| ThaiBMA | Reference prices, THOR, market statistics | Daily/Real-time | Mark-to-market, benchmark comparison |
| Bloomberg/Reuters | Real-time quotes, analytics, news | Real-time | Pre-trade analysis, price validation |
| BOT | Policy rate, yield curves, auction announcements | As published | Monetary policy operations, pricing |
| Internal | Historical trades, counterparty quotes | Continuous | Relationship pricing, best execution |

#### 3.1.5 ThaiBMA Trade Reporting Interface (30-Minute Reporting Obligation) **Automated reporting workflow**:

```
Trade Execution → Immediate Capture → Validation → Formatting →
Transmission → Acknowledgment Receipt → Exception Handling (if needed)
```

| Monitoring Element | Threshold | Action |
|---|---|---|
| Time to report | 20 minutes (10-min buffer) | Amber alert |
| Time to report | 28 minutes | Red alert, manual escalation |
| Acknowledgment pending | 5 minutes | Retry, backup transmission |
| Rejection received | Immediate | Investigation, correction, resubmission |

### 3.2 Settlement Module

**3.2.1 TSD Integration for Corporate Bond Settlement (DVP Model 1)** **TSD DVP Model 1** achieves **simultaneous, irrevocable settlement** of securities and cash on **gross, trade-by-trade basis**. **Critical integration points**:

| Message Type | Purpose | Timing |
|---|---|---|
| Settlement instruction | Initiate DVP process | T+0 to T+2, before cutoff |
| PSMS submission | Pre-settlement matching | Prior to settlement date |
| Position inquiry | Verify available securities | Pre-trade, pre-settlement |
| Corporate action notification | Process coupons, redemptions | As announced |
| Confirmation | Verify settlement completion | Real-time upon execution |

**Since November 2017: TSD prohibits debit balances**—trades with insufficient securities remain pending until resolved.

**3.2.2 Bank of Thailand Settlement for Government Securities (BAHTNET RTGS)** **BAHTNET RTGS** provides **immediate finality** for government securities and high-value payments. Integration requirements:

| Function | Implementation |
|---|---|
| Payment message generation | SWIFT MT103 or domestic equivalent, unique references |
| Liquidity monitoring | Real-time central bank reserve balance, projected needs |
| Settlement prioritization | Time-critical payments first, queue management |
| Finality confirmation | Irrevocable settlement acknowledgment |
| End-of-day reconciliation | BAHTNET statement vs. internal records |

**3.2.3 Repo Lifecycle Management**

| Lifecycle Stage | System Functionality | Key Data Elements |
|---|---|---|
| **Initial trade** | Record terms, select collateral, generate instructions | Rate, tenor, collateral ID, haircut, margin terms |
| **Daily operations** | Mark-to-market, margin calculation, substitution processing | Market prices, exposure vs. collateral value, margin calls |
| **Coupon processing** | Pass-through to collateral provider (if applicable) | Coupon amount, timing, tax treatment |
| **Maturity** | Calculate final amount, prepare settlement, close position | Principal + accrued interest, return collateral |
| **Rollover** | Terminate original, create replacement, document continuity | New rate, tenor, collateral (same or refreshed) |

**3.2.4 Pre-Settlement Matching System (PSMS) Interface** **PSMS mandatory matching fields -** | **Field** | **Source** | **Validation** | |:—|:—|:—| | Trade type (DvP/FoP) | Trade capture | System default DvP for purchases | | Settlement amount | Calculated | Matches trade economics | | Settlement date | Calculated | Business day, within T+2 | | Securities volume | Trade capture | Matches nominal amount | | ISIN | Security master | Valid, active | | Counterparty depository account | Counterparty master | Mandatory for bonds |

**3.2.5 Settlement Failure Handling and Exception Management**

| Failure Type | Root Cause | Resolution Workflow |
|---|---|---|
| Securities fail | Insufficient position | Securities borrowing, purchase, or settlement date extension |
| Cash fail | Insufficient BAHTNET funds | Intraday liquidity mobilization, overdraft arrangement |
| Technical fail | System/communication error | IT escalation, manual processing contingency |
| Mismatch fail | PSMS rejection | Counterparty coordination, correction, resubmission |

**3.2.6 T+2 Settlement Cycle Compliance** **Settlement date calculation** must incorporate:

- **Thai business day convention** (Monday–Friday, excluding holidays)

- **Official holiday calendar** (Bank of Thailand published)
- **Cutoff time awareness** (TSD: typically 14:00 for same-day; BAHTNET: session-based)

### 3.3 Accounting Module

#### 3.3.1 TFRS 9 / IFRS 9 Financial Instrument Classification

| Category | Business Model | SPPI Test | Measurement | P&L Impact |
|---|---|---|---|---|
| **Amortized Cost** | Hold to collect | Pass | Amortized cost using EIR | Interest income only; impairment if any |
| **FVOCI** | Hold to collect and sell | Pass | Fair value; OCI for FV changes | Interest income in P&L; FV changes in OCI, recycled on sale |
| **FVTPL** | Trading or other | Fail or elective | Fair value | All changes in P&L |

**3.3.1.1 Amortized Cost Category (Hold-to-Collect Business Model)** **Appropriate for**: Core liquidity buffers, strategic government bond holdings, high-grade corporate bonds with long-term hold intent.

**System implementation**:

- **Effective interest rate calculation** at initial recognition:

$$EIR = \text{rate that discounts estimated future cash flows to initial carrying amount}$$

- **Premium/discount amortization**: systematic over instrument life to maturity
- **Impairment**: 12-month ECL (Stage 1) or lifetime ECL if credit deterioration

**3.3.1.2 Fair Value Through Other Comprehensive Income (FVOCI)** **Appropriate for**: Flexible liquidity portfolios where sale is possible but not primary objective.

**Critical mechanics**:

- **Dual tracking**: amortized cost basis for interest income; fair value for balance sheet
- **OCI recycling**: cumulative FV gains/losses transfer to P&L upon derecognition
- **Impairment**: recognized in P&L (not OCI), with OCI balance adjustment

**3.3.1.3 Fair Value Through Profit or Loss (FVTPL)** **Appropriate for**: Trading positions, instruments failing SPPI (e.g., convertible bonds), elective designation to eliminate accounting mismatch.

**Characteristics**:

- **Transaction costs expensed immediately** (not capitalized)
- **No impairment**—all credit risk reflected in fair value
- **Most transparent** for trading performance, highest P&L volatility

**3.3.2 SPPI Test Implementation for Instrument Eligibility** **SPPI failure indicators** requiring manual assessment:

| Feature | SPPI Impact | Example |
|---|---|---|
| Leverage | Fail | Inverse floater, leveraged return |
| Non-standard interest | Fail | Equity-linked, commodity-linked coupon |
| Prepayment option (favorable) | Potential fail | Deep discount prepayment at par |
| Extension option (non-market) | Potential fail | Extension at below-market rate |
| Conversion feature | Fail | Convertible bonds |

**3.3.3 Expected Credit Loss (ECL) Impairment Calculation**

| Component | Data Source | Methodology |
|---|---|---|
| Probability of Default (PD) | Internal models, rating agencies | Point-in-time or through-the-cycle, forward-looking |
| Loss Given Default (LGD) | Historical recovery, collateral | Workout LGD or market LGD |
| Exposure at Default (EAD) | Amortized cost, expected drawdowns | Current exposure + expected changes |
| Discount rate | Effective interest rate | Original EIR or approximation |

**Macroeconomic integration**: Thai GDP growth, unemployment, industry-specific indicators forward-looking through **reasonable and supportable** forecast period.

**3.3.4 Repo Accounting Treatment (Securities Remain on Borrower's Balance Sheet)** **Fundamental principle**: Economic substance as **secured financing**, not sale.

| Party | Accounting Treatment |
|---|---|
| **Cash borrower (repo seller)** | Securities remain assets; recognize cash liability; accrue repo interest expense |
| **Cash lender (reverse repo buyer)** | Cash derecognized; recognize receivable; do not recognize collateral securities (unless rehypothecation criteria met) |

### 3.3.5 Accrued Interest and Repo Fee Recognition

| Instrument Type | Accrual Basis | Day Count | Recognition |
|---|---|---|---|
| Fixed-rate bonds | Time-proportion | Actual/365 or Actual/Actual | Effective interest method |
| Floating-rate bonds | From reset date | Actual/365 | Contractual rate + margin |
| T-Bills (discount) | Accretion | Actual/365 | Yield to maturity |
| Repos | From value date | Actual/365 | Repo rate on cash amount |
| Reverse repos | From value date | Actual/365 | Repo rate on cash advanced |

### 3.3.6 Mark-to-Market Valuation and ThaiBMA Price Integration

| Valuation Hierarchy | Input Type | Thai Market Application |
|---|---|---|
| Level 1 | Quoted prices in active markets | ThaiBMA reference prices for liquid government bonds |
| Level 2 | Observable inputs, direct or indirect | Matrix pricing for corporate bonds, dealer quotes |
| Level 3 | Unobservable inputs | Internal models for illiquid positions, significant adjustments |

**ThaiBMA price integration**: Automated daily retrieval, validation against movement thresholds (e.g., ±5% from previous day triggers review), fallback to alternative sources if unavailable.

### 3.3.7 Journal Entry Generation and General Ledger Posting

| Transaction/Event | Entry Type | Key Accounts |
|---|---|---|
| Bond purchase (amortized cost) | Trade date | Dr Investment, Cr Payable |
| Bond purchase settlement | Settlement date | Dr Payable, Cr Cash |
| Interest accrual | Periodic | Dr Accrued interest receivable, Cr Interest income |
| Coupon receipt | Cash flow | Dr Cash, Cr Accrued interest receivable |
| FVOCI fair value change | Period-end | Dr/Cr FVOCI reserve (OCI), Cr/Dr Investment |
| FVOCI sale | Derecognition | Dr Cash, Cr Investment; recycle FVOCI to P&L |
| Repo initiation | Trade date | Dr Cash, Cr Repo payable; securities remain |
| Repo interest accrual | Periodic | Dr Repo interest expense, Cr Accrued repo payable |

## 3.4 Reporting Module

### 3.4.1 Regulatory Reporting

#### 3.4.1.1 ThaiBMA Transaction and Position Reports

| Report Type | Frequency | Content | Format |
|---|---|---|---|
| Transaction report | Real-time (30-min) | Trade details, price, volume, counterparty | ThaiBMA specified API |
| Position report | Daily/Periodic | Holdings by ISIN, maturity, rating | Aggregated, confidential |
| Market activity | As required | Turnover, market share analysis | Statistical |

#### 3.4.1.2 Bank of Thailand Liquidity and Statistical Returns

| Return Type | Purpose | Key Data Elements |
|---|---|---|
| LCR calculation | Liquidity adequacy | HQLA classification, cash outflows, inflows |
| NSFR calculation | Structural funding stability | Available stable funding, required stable funding |
| Monetary policy operations | Central bank oversight | Repo participation, collateral usage, rates |
| Securities statistics | Market monitoring | Holdings by type, maturity, counterparty |

**HQLA classification -**

| **Asset Class** | **LCR Level** | **Haircut** | **System Tracking** |
|:---|:---|:---|:---|
| Government bonds (THB) | Level 1 | 0% | Automatic eligibility |
| BOT bonds/bills (THB) | Level 1 | 0% | Automatic eligibility |
| SOE bonds (guaranteed, rated AA- or above) | Level 1 | 0% | Guarantee verification |
| Corporate bonds (rated AA- or above) | Level 2A | 15% | Rating monitoring, cap tracking |

#### 3.4.1.3 SEC Disclosure Requirements for Corporate Bond Holdings

| Trigger | Disclosure | Timeline | System Support |
|---|---|---|---|
| >5% of class/series | Holding notification | Within specified period | Automated threshold monitoring |
| >25% of issued amount | Significant holding | Immediate | Alert generation, form preparation |
| Related-party transaction | Enhanced disclosure | As required | Counterparty relationship flagging |

### 3.4.2 Management Reporting

#### 3.4.2.1 Portfolio Valuation and Performance Attribution

| Metric | Calculation | Use |
|---|---|---|
| Total return | Income + price change + FX (N/A for THB) | Overall performance |
| Excess return vs. benchmark | Portfolio return − benchmark return | Active management value-add |

| Metric | Calculation | Use |
|---|---|---|
| Duration contribution | Position DV01 × yield change | Interest rate risk attribution |
| Credit spread contribution | Spread change × spread duration | Credit risk attribution |
| Security selection | Actual return − model return | Picking skill assessment |

#### 3.4.2.2 Liquidity Position and Cash Flow Forecasting

| Forecast Horizon | Purpose | Key Inputs |
|---|---|---|
| Intraday | BAHTNET funding management | Real-time positions, pending settlements |
| 1–7 days | Operational liquidity | Maturing repos, known cash flows |
| 1–3 months | Tactical planning | Bond maturities, seasonal patterns |
| 1 year+ | Strategic ALM | Bond portfolio cash flows, liability matching |

#### 3.4.2.3 Risk Exposure Dashboards

| Risk Category | Metrics | Visualization |
|---|---|---|
| Market risk | DV01, duration, VaR, stress test results | Heat maps, trend charts, scenario comparison |
| Credit risk | Exposure by rating, sector, issuer; ECL trends | Concentration pie charts, migration matrices |
| Liquidity risk | LCR, NSFR, survival horizon, funding gaps | Time-series, stress scenario waterfalls |
| Operational risk | Settlement fails, confirmation delays, system outages | Exception dashboards, root cause tracking |

### 3.4.3 Audit Trail and Transaction History

| Retention Requirement | Implementation |
| --- | --- |
| 5 years minimum (Thai regulations) | Online accessible for 2 years, archival storage thereafter |
| Immutable logging | Write-once storage, cryptographic hashing, or blockchain |
| Comprehensive capture | All user actions, system processes, external interfaces |
| Searchable, reconstructible | Point-in-time position reconstruction, transaction lifecycle tracing |

### 3.5 Compliance and Risk Control Module

#### 3.5.1 Counterparty Credit Limit Management

| Limit Dimension | Measurement | Monitoring Frequency |
| --- | --- | --- |
| Single name | Sum of all exposures (direct + contingent) | Real-time |
| Group/connected | Aggregated across legal entities | Daily |
| Tenor bucket | Time-banded exposure profile | Daily |
| Collateral-adjusted | Exposure net of haircut-adjusted collateral | Real-time for repos |

#### 3.5.2 Concentration Risk Monitoring (Single Issuer, Sector, Rating)

| Concentration Type | Typical Limit | System Action |
| --- | --- | --- |
| Single issuer | 10–15% of portfolio | Warning at 80%, escalation at 100% |
| Sector (e.g., financials) | 25–30% of portfolio | Board-level reporting if approached |
| Rating category (e.g., BBB) | 20% of corporate exposure | Stress testing, downgrade scenario analysis |
| Sovereign/quasi-sovereign | Regulatory large exposure limits | Hard regulatory caps |

#### 3.5.3 Market Risk Measurement (Duration, DV01, VaR)

| Metric | Application | Calculation Method |
| --- | --- | --- |
| Modified duration | Interest rate sensitivity | Price change for 1% yield shift |
| DV01 | Monetary risk exposure | Price change for 1bp yield shift |
| Convexity | Second-order rate risk | Adjustment for large yield moves |
| VaR (95% or 99%) | Potential loss estimate | Historical simulation using Thai market data |
| Stress testing | Extreme scenario impact | Parallel shift, steepening, credit spread widening |

#### 3.5.4 AML/KYC Integration and Suspicious Transaction Monitoring

| Integration Point | Functionality |
| --- | --- |
| Counterparty onboarding | KYC verification, sanctions screening, PEP check |
| Transaction monitoring | Pattern analysis, threshold alerts, unusual activity flags |
| Regulatory reporting | STR (Suspicious Transaction Report) generation for AMLO |
| Record keeping | 5-year retention, audit trail for examinations |

#### 3.5.5 Four-Eyes Principle and Trade Approval Workflows

| Transaction Characteristic | Approval Requirement |
| --- | --- |
| Standard, within limits | Trader + confirmation officer |
| Near limit (80–100%) | Additional risk manager approval |
| Limit breach (with override) | Senior management, documented justification |
| Non-standard terms | Product control, legal review |
| New counterparty | Compliance, credit approval |

#### 3.5.6 Regulatory Capital Adequacy Reporting (BOT Requirements)

| Risk Type | Approach | System Support |
|---|---|---|
| Credit risk (standardized) | Risk weights by rating, collateral recognition | Automated RWA calculation, exposure aggregation |
| Market risk (trading book) | Standardized or internal model | VaR backtesting, model validation data |
| Operational risk | Basic indicator or standardized approach | Loss data collection, scenario analysis |

## 4. User Role Architecture and Access Control

### 4.1 Front Office Functions

#### 4.1.1 Trader/Dealer Role (Trade Execution, Price Quoting)

| Permission Category | Specific Capabilities | Explicit Restrictions |
|---|---|---|
| Market data | Real-time prices, yields, analytics, news | No historical data export beyond role needs |
| Trade capture | Deal entry, modification (pre-confirmation), cancellation | No post-confirmation modification; no settlement instruction change |
| Position inquiry | Real-time P&L, exposure, limit utilization | No risk parameter modification |
| Counterparty interaction | Quote generation, negotiation logging | No master agreement terms modification |

#### 4.1.2 Portfolio Manager Role (Investment Strategy, Position Monitoring)

| Function | System Support |
|---|---|
| Strategic allocation | Benchmark selection, target setting, rebalancing analysis |
| Performance analysis | Attribution, peer comparison, trend identification |
| Scenario modeling | What-if analysis, stress testing, optimization |

| Function | System Support |
|---|---|
| Reporting | Management dashboards, board presentations |

#### 4.1.3 Market Access and Real-Time Data Permissions

| Data Service | Typical Front Office Access | Cost Management |
|---|---|---|
| ThaiBMA | Full—reference prices, THOR, market stats | Core infrastructure, unlimited |
| Bloomberg Terminal | Trading, analytics, messaging | Limited licenses, shared if needed |
| Reuters Eikon | Alternative/supplementary | Selective based on trading focus |
| Market data feeds | Real-time prices for trading systems | Exchange fees, managed by usage |

### 4.2 Middle Office Functions

#### 4.2.1 Risk Manager Role (Limit Setting, Risk Monitoring, VaR Calculation)

| Core Responsibility | System Capability Required |
|---|---|
| Limit framework maintenance | Parameter configuration, hierarchy management, approval workflows |
| Real-time monitoring | Dashboards, alerts, exception queues |
| VaR/model operation | Model execution, backtesting, validation support |
| Stress testing | Scenario design, execution, reporting |
| Regulatory capital | RWA calculation, ICAAP support |

#### 4.2.2 Compliance Officer Role (Regulatory Reporting, Policy Enforcement)

| Function | System Support |
|---|---|
| Regulatory reporting | Automated generation, validation, submission tracking |

| Function | System Support |
|---|---|
| Policy rule engine | Configurable controls, automated checking |
| Monitoring surveillance | Pattern detection, unusual activity identification |
| Examination support | Audit trail retrieval, documentation preparation |

#### 4.2.3 Product Control Role (P&L Attribution, Independent Price Verification)

| Activity | Methodology | System Implementation |
|---|---|---|
| Daily P&L calculation | Trade-based, position-based reconciliation | Automated with exception investigation |
| Independent price verification | Comparison to ThaiBMA, broker quotes, models | Threshold-based escalation, documentation |
| Reserve calculation | Prudent valuation adjustments | Model-based with management judgment |
| Attribution analysis | Return decomposition by risk factor | Automated reporting, drill-down capability |

### 4.3 Back Office Functions

#### 4.3.1 Settlement Officer Role (Trade Confirmation, Settlement Instruction Generation)

| Process Stage | System Functionality | Key Output |
|---|---|---|
| Trade confirmation | Matching with counterparty, discrepancy resolution | Matched confirmation record |
| Instruction generation | TSD/BAHTNET message creation, validation | Formatted, authorized instructions |
| Transmission | Secure sending, acknowledgment monitoring | Sent status, confirmation receipt |
| Settlement monitoring | Status tracking, fail identification, resolution | Completed settlement, or exception queue |

#### 4.3.2 Accounting Officer Role (Journal Entry Review, Reconciliation)

| Responsibility | System Support |
|---|---|
| Entry review | Automated generation, manual approval workflow, exception handling |
| GL posting | Controlled interface, validation, error correction |
| Reconciliation | Nostro, depot, intersystem reconciliation with automated matching |
| Period-end close | Accrual processing, valuation, reporting preparation |

#### 4.3.3 Operations Manager Role (Exception Handling, Failed Trade Resolution)

| Management Function | System Tool |
|---|---|
| Exception queue oversight | Dashboard with aging, categorization, assignment |
| Resource allocation | Workload balancing, priority setting |
| Process improvement | Metrics analysis, trend identification, root cause tracking |
| Escalation management | Defined thresholds, automatic routing, senior notification |

### 4.4 IT Administration Functions

#### 4.4.1 System Administrator Role (User Provisioning, Security Configuration)

| Administrative Function | Control Safeguard |
|---|---|
| User account lifecycle | Dual authorization for creation, modification, deletion |
| Role assignment | Segregation of duties validation, periodic recertification |
| Password policy | Complexity, rotation, lockout configuration |
| Security monitoring | Access logging, anomaly detection, incident response |

#### 4.4.2 Data Manager Role (Static Data Maintenance, Instrument Setup)

| Data Category | Maintenance Activity | Quality Control |
|---|---|---|
| Security master | New instrument setup, terms updates, status changes | Validation against authoritative sources, dual verification |
| Counterparty master | Onboarding, limit setup, agreement linkage | KYC integration, credit approval workflow |
| Market data | Price source configuration, holiday calendar, benchmark curves | Automated validation, manual override logging |
| Reference data | Ratings, corporate actions, regulatory classifications | Source verification, change notification |

#### 4.4.3 Interface Manager Role (External System Connectivity, Message Monitoring)

| Interface | Monitoring Element | Response Procedure |
|---|---|---|
| TSD | Message flow, settlement status, position reconciliation | Alert on delay, investigation, escalation |
| BAHTNET | Payment status, liquidity position, statement reconciliation | Intraday liquidity management, fail resolution |
| ThaiBMA | Trade reporting confirmation, price feed availability | Retry, backup, manual submission if needed |
| Market data vendors | Feed status, data quality, timeliness | Fallback activation, vendor escalation |

## 5. External System Integration Architecture

### 5.1 Market Infrastructure Connections

#### 5.1.1 Thailand Securities Depository (TSD) — SWIFT/Proprietary Messaging

| Integration Layer | Protocol | Message Types | Criticality |
|---|---|---|---|
| Connectivity | VPN/leased line with SWIFT or proprietary | MT540–548 (settlement), custom (inquiry, corporate actions) | High—settlement failure disrupts operations |
| Security | PKI encryption, digital signatures | All messages authenticated | Regulatory requirement |
| Resilience | Dual connectivity, automatic failover | Message queuing, guaranteed delivery | Business continuity essential |

#### 5.1.2 Bank of Thailand BAHTNET — RTGS Payment and Settlement

| Feature | Specification | System Implication |
|---|---|---|
| Settlement mode | Real-time gross settlement | Immediate finality, no batch processing |
| Operating hours | Defined session, intraday finality | Liquidity management within session constraints |
| Message format | SWIFT MT or domestic equivalent | Format transformation, validation |
| Liquidity access | Intraday repo, standing facilities | Integration with monetary policy operations |

#### 5.1.3 Thai Bond Market Association (ThaiBMA) — Trade Reporting API

| Service | Integration Mode | Performance Requirement |
|---|---|---|
| Real-time trade reporting | API, XML/JSON format | <30 minutes from execution |
| Reference price retrieval | Scheduled batch, on-demand query | Daily for valuation, intraday if available |
| Market statistics | Periodic download, subscription | Daily/weekly for analysis |
| THOR publication | Automated ingestion | Daily fixing for benchmark use |

### 5.2 Market Data and Pricing Feeds

**5.2.1 ThaiBMA Reference Pricing for Mark-to-Market** **Primary valuation source** for Thai bonds, with **hierarchical fallback**:

| Priority | Source | Application |
|---|---|---|
| 1 | ThaiBMA reference price | Standard mark-to-market |
| 2 | ThaiBMA indicative price | Illiquid instruments |
| 3 | Broker quotes (2+ sources) | No ThaiBMA price available |
| 4 | Model valuation (documented) | No observable market data |

#### 5.2.2 Bloomberg/Reuters Integration for Market Data

| Use Case | Typical Implementation | Cost Consideration |
|---|---|---|
| Real-time trading | Bloomberg Terminal, limited licenses | ~$20,000+ per terminal annually |
| Analytics and research | Bloomberg/Reuters desktop, shared access | License tier based on user count |
| Automated feeds | API integration for selected data | Per-field or bundle pricing |
| News and alerts | Integrated with trading workflow | Often bundled with price data |

#### 5.2.3 BOT Monetary Policy Rate and Benchmark Curve Data

| Data Element | Source | Frequency | Use |
|---|---|---|---|
| Policy rate | BOT announcement | As decided (typically quarterly) | Pricing anchor, expectation gauge |
| THOR | ThaiBMA calculation | Daily | Overnight repo benchmark |
| Government yield curve | BOT/ThaiBMA publication | Daily | Relative value, discounting |
| Repo rates by tenor | ThaiBMA average | Daily | Repo pricing reference |

### 5.3 Internal System Integration

#### 5.3.1 Core Banking System Interface (General Ledger, Customer Accounts)

| Integration Mode | Data Exchange | Control Consideration |
|---|---|---|
| Real-time API | Journal entries, position updates | Immediate posting, reconciliation challenge |
| Batch file exchange | End-of-day summary, detailed transactions | Controlled timing, validation before posting |
| Message queue | Event-driven updates, guaranteed delivery | Balance between timeliness and control |

#### 5.3.2 Enterprise Risk Management System Connectivity

| Risk Data Flow | Content | Aggregation Level |
|---|---|---|
| To ERM | Position, exposure, limit utilization | Detailed, instrument-level |
| From ERM | Enterprise risk metrics, stress scenarios | Consolidated, portfolio-level |
| Bi-directional | Model parameters, validation results | As required for consistency |

#### 5.3.3 Data Warehouse and Business Intelligence Layer

| Warehouse Function | Implementation Approach |
|---|---|
| Historical data retention | 10+ years online, archival for regulatory |
| Analytics and reporting | Star schema, dimensional modeling |
| Data quality | ETL validation, reconciliation, exception handling |
| Access layer | Self-service BI for authorized users, governed data marts |

## 6. Regulatory Compliance Framework

### 6.1 Bank of Thailand Supervisory Requirements

#### 6.1.1 Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR) Reporting

| Ratio | Numerator | Denominator | Minimum Requirement |
|---|---|---|---|
| LCR | High-quality liquid assets (HQLA) | Net cash outflows (30-day stress) | 100% |
| NSFR | Available stable funding | Required stable funding | 100% |

**System support**: Automated HQLA classification, cash flow projection, stress scenario modeling, regulatory report generation.

#### 6.1.2 Large Exposure Limits and Concentration Risk Controls

| Exposure Type | Limit Basis | System Tracking |
|---|---|---|
| Single counterparty | Tier 1 capital percentage | Real-time aggregation across products |
| Connected group | Consolidated group definition | Relationship mapping, attribution |
| Sector concentration | Internal risk appetite | Industry classification, monitoring |
| Sovereign concentration | Regulatory and internal limits | Government, SOE, guaranteed aggregation |

#### 6.1.3 Primary Dealer Obligations (if applicable)

| Obligation | System Support Required |
|---|---|
| Auction participation | Bid preparation, submission, allocation processing |
| Market making | Quote generation, distribution, performance tracking |
| Information provision | Yield quotations, market data submission to BOT/ThaiBMA |
| Inventory management | Position optimization, regulatory reporting |

### 6.2 Securities and Exchange Commission (SEC) Requirements

#### 6.2.1 Corporate Bond Disclosure and Transparency Obligations

| Disclosure Trigger | Requirement | System Function |
| --- | --- | --- |
| Material holding threshold | Notification to issuer, SEC | Automated threshold monitoring, form generation |
| Related-party transaction | Enhanced disclosure, board approval | Counterparty relationship flagging, workflow |
| Inside information | Restricted trading, disclosure | Information barrier management, watch lists |

#### 6.2.2 Investor Protection and Fair Dealing Standards

| Standard | Implementation |
| --- | --- |
| Best execution | Order routing analysis, price improvement documentation |
| Suitability | Customer classification, product appropriateness (if customer-facing) |
| Conflicts of interest | Chinese walls, personal account dealing monitoring |

### 6.3 ThaiBMA Market Conduct Rules

#### 6.3.1 Market Making and Quoting Obligations

| Rule Element | Compliance Approach |
| --- | --- |
| Two-way quoting | Automated quote generation, monitoring, escalation if unable |
| Quote firmness | Time-stamped, honored within defined parameters |
| Spread maintenance | Within market convention, documented exceptions |

#### 6.3.2 Trade Reporting Accuracy and Timeliness

| Requirement | System Implementation |
|---|---|
| 30-minute reporting | Automated capture, validation, transmission with monitoring |
| Complete and accurate | Field validation, counterparty confirmation matching |
| Correction handling | Amendment workflow, audit trail, re-notification |

### 6.4 Tax Compliance

#### 6.4.1 Withholding Tax Calculation on Interest Income

| Income Type | Withholding Rate | System Function |
|---|---|---|
| Government bond interest | Exempt or reduced per specific bond | Tax status tracking, exemption certificate |
| Corporate bond interest | 15% standard (individuals); 1% juristic persons | Automated calculation, withholding, reporting |
| Foreign investor | Treaty rate or 15% | Tax residency verification, treaty claim support |

#### 6.4.2 Specific Business Tax (SBT) on Bond-Related Income

| Activity | SBT Rate | Calculation Basis |
|---|---|---|
| Bond trading (banking license) | 0.01% | Gross profit before expenses |
| Interest, discounts, fees | 0.01% | Gross amount |

**System support**: SBT accrual, return preparation, reconciliation with corporate income tax.

#### 6.4.3 Repo Transaction Tax Treatment (Interest vs. Capital Gains)

| Tax Characterization | Treatment | System Implementation |
|---|---|---|
| Repo interest/fee | Interest income/expense, ordinary rates | Accrual tracking, tax reporting |

| Tax Characterization | Treatment | System Implementation |
|---|---|---|
| Manufactured payments | Pass-through, potential withholding | Tracking, documentation, reclaim |
| Capital gains on collateral | Not recognized (economic substance as financing) | Position tracking, gain/loss suppression |

## 7. Technical Architecture Considerations

### 7.1 System Scalability and Performance

#### 7.1.1 Low-Volume Optimization ( 25 T/day Design Point)

| Design Decision | Low-Volume Optimization | Future-Proofing |
|---|---|---|
| Database | Single-node RDBMS, full normalization | Partitioning strategy for growth |
| Application server | Single primary with warm standby | Horizontal scaling path if needed |
| Batch processing | Generous windows, thorough validation | Job scheduling, resource management |
| User interface | Rich, contextual, moderate latency | Responsive design, progressive enhancement |

#### 7.1.2 Modular Component Design for Future Expansion

| Module | Current Scope | Expansion Path |
|---|---|---|
| Trade execution | THB bonds, repos | FX, derivatives, multi-currency |
| Settlement | TSD, BAHTNET | Cross-border, CLS, DLT exploration |
| Accounting | TFRS 9 | IFRS 17, new standards adoption |
| Reporting | Thai regulatory | Global regulatory, ESG integration |

### 7.2 Security and Controls

#### 7.2.1 Role-Based Access Control (RBAC) Implementation

| RBAC Layer | Enforcement Mechanism |
|---|---|
| Database | Row-level security, column encryption for sensitive data |
| Application | Function-level permissions, menu filtering, field-level access |
| Network | VLAN segmentation, VPN for remote access, intrusion detection |
| Physical | Data center access controls, environmental security |

#### 7.2.2 Data Encryption and Secure Messaging

| Data State | Protection |
|---|---|
| At rest | AES-256 encryption, key management (HSM) |
| In transit | TLS 1.3, certificate pinning for critical interfaces |
| In use | Memory encryption, secure enclaves for sensitive operations |

#### 7.2.3 Disaster Recovery and Business Continuity Planning

| Scenario | Recovery Objective | Implementation |
|---|---|---|
| System failure | RTO: 4 hours, RPO: 1 hour | Hot standby, automatic failover |
| Site disaster | RTO: 24 hours, RPO: 4 hours | DR site, replication, runbook |
| Regional event | RTO: 72 hours, RPO: 24 hours | Alternative region, backup restoration |

### 7.3 Audit and Logging

#### 7.3.1 Comprehensive Transaction Logging (5-Year Retention per Thai Regulations)

| Log Category | Content | Retention |
|---|---|---|
| User access | Login/logout, failed attempts, privilege escalation | 5+ years |

| Log Category | Content | Retention |
| --- | --- | --- |
| Data modification | Before/after values, user, timestamp, approval chain | 5+ years |
| Transaction lifecycle | All states, system actions, external communications | 5+ years |
| System events | Configuration changes, errors, performance metrics | 3+ years operational, 5+ years summary |

#### 7.3.2 Immutable Audit Trail for Regulatory Examination

| Immutability Mechanism | Implementation | Verification |
| --- | --- | --- |
| Write-once media | Optical storage for archival | Physical media verification |
| Cryptographic hashing | Blockchain or Merkle tree structure | Hash chain validation |
| Third-party attestation | External audit, continuous monitoring | Independent verification reports |